
Re: Data Coverage on Finstream Server for 10-K/10-Q and Earnings Calls

From Sarthak Vishnu <S.G.Vishnu@sms.ed.ac.uk>

Date Wed 2026-06-17 18:22

To Tiejun Ma <tiejun.ma@ed.ac.uk>

Cc Sunnie Li <sunnie.y.li@ed.ac.uk>

 1 attachment (3 KB)

references.txt;

Dear Prof Ma,

I hope you are well. I am writing to share a progress update and to seek your guidance on a few design decisions.

The following datasets have been collected and indexed:

- Full SEC filings (10-K and 10-Q): 2,285 classified filings for 634 CIKs, 2025–2026 (from FINSTREAM database)
- Earnings call transcripts: 299 calls, 172 tickers, 2025–2026 (from FINSTREAM database)
- Form 4 insider transactions: 1,670,042 non-derivative transactions, 645 companies, 2006–2026, collected from SEC EDGAR quarterly flat files
- GDELT Events Database: two-stage collection, described below

GDELT Collection:

Two collection approaches were pursued in parallel:

- **Approach 1 — Direct Download (GDELT 1.0 GKG):** GDELT 1.0 publishes one file per calendar day. I downloaded (manually using a script) and filtered for 2025–2026, matching articles to S&P 500 companies. Output columns include avg_tone, avg_tone_neg, n_articles, n_sources, avg_polarity, and finance-specific theme flags.
- **Approach 2 — Google BigQuery (GDELT 2.0 GKG):** GDELT 2.0 publishes one file every 15 minutes (~50,000+ files per year), making direct download entirely impractical. Google BigQuery hosts the full dataset as a single quarriable table returning only the relevant rows. I have created a GCP project for enabled the BigQuery API. The only remaining step is credit card registration for identity verification — Google requires a linked billing account even for free-tier queries, though estimated query cost for our scope is [for full 2015–2026 coverage: ~\$15] along with the 1 TB/month free allowance. If activated, it would also allow extension to historical coverage from 2015 onwards.

NOTE: GDELT GKG 2.0 exists from Feb 2015, and for collecting data for Apr 2013 to Feb 2015 we will still have to rely on GDELT GKG1.0. There's no GKG data for period before Apr 2013.

Dataset Integration — Which Datasets Belong in NLP Training vs Ablation

The key question for each dataset is whether it contributes text or signals that improve DAPT, Contrastive Fine-tuning, or BERTopic, or whether its value is purely as a scalar predictor in Stage 5 Ablation. I evaluated each dataset against this criterion using the literature survey below.

Full 10-K (beyond Item 1A and Item 7A): excluded from NLP training.

Campbell et al. (2014, Review of Accounting Studies) show that Item 1A accounts for approximately 11% of 10-K words but carries the specific market-risk and idiosyncratic-risk signal. The remaining 89% consists of business description (marketing prose), financial statements (tabular, non-linguistic), legal exhibits, and boilerplate certification text. Adding these sections would introduce vocabulary that dilutes the risk-disclosure specialisation that is the core contribution of the pipeline. Magner et al. (2025, Finance Research Letters) achieve their best predictive results using only Item 1A and they do not use the full 10-K. These two papers together provide the decisive justification for retaining Item 1A as the primary 10-K text source.

10-Q (quarterly filings): excluded from NLP training.

The quarterly 10-Q does not contain Item 1A in the same mandatory form as the annual 10-K. Its "Risk Factors" update section typically states that risks have not materially changed from the most recent Annual Report — it is largely boilerplate with minimal new risk-language content. There is no precedent in the literature for quarterly filings in risk-embedding pre-training.

Form 4 (insider transactions): excluded from NLP training.

Recommended for Ablation as scalar features. Form 4 is purely tabular: transaction dates, share counts, prices, and officer roles. There is no language for a masked language model (MLM) to train on and no text from which to construct contrastive pairs. The signal value, however, is real and well-evidenced. Cohen, Malloy & Pomorski (2012, Journal of Finance) show that trades by "opportunistic" insiders — those whose trade timing is not explained by a mechanical calendar pattern — produce 82 bps/month abnormal return, while "routine" traders show near-zero alpha. Goldie, Jiang, Koch & Wintoki (2023, JFQA) further identify that indirect trades (through family, trust, or retirement accounts, reported via Form 4 Field 7) carry 89 bps/month, exceeding direct trades at 65 bps/month. Both signals are constructible from the collected data and are recommended as scalar features in the Ablation.

GDELT : excluded from NLP training

Recommended for Ablation as scalar features. A critical constraint is that only aggregated sentiment scores were collected such as `avg_tone`, `n_articles`, `n_sources`, `avg_polarity` and not the raw article text. The output contains scalars, not sentences, and a language model has nothing to train on from this format. If raw article text were available via the `source_url` field in the raw GKG files, adding news articles to the DAPT corpus would be theoretically defensible. Boudoukh, Feldman, Kogan & Richardson (2019, Review of Financial Studies) show firm-specific news drives approximately 50% of overnight idiosyncratic return variance, suggesting a news-aware encoder would improve risk language understanding. However, downloading, deduplicating, and cleaning full article text from over 4 million GDELT article-company rows is a substantial task which would take more time.

Earnings Calls: Recommended for NLP Training at Two Levels

This is the only new dataset with a scientific case for integration into the NLP training phases, not just Ablation.

- *DAPT corpus expansion:* Earnings calls share the financial risk register with Item 1A: executives use the same vocabulary ("headwinds", "exposure", "uncertainty", "liability") and the analyst Q&A section in particular mirrors the risk probing that Item 1A disclosures address. Published precedents: Fin-E5 (Tang & Yang, EMNLP 2025) trains on 64 financial datasets explicitly including earnings transcripts and achieves state-of-the-art on FinMTEB; FinGPT (Yang et al. 2023) trains on SEC filings and earnings calls jointly. Gururangan et al. (2020, ACL) establish that DAPT on any domain-relevant text consistently improves downstream task performance.

- *A 4th contrastive view (proposed novel contribution)*: The IPP proposes three contrastive views. I propose a cross-document risk view as a 4th:

View 1 – Lexical [unchanged]: overlapping spans, same Item 1A

View 2 – Chronological [unchanged]: same firm, different fiscal years

View 3 – Sector [unchanged]: same SIC code, different firm [unchanged]

View 4 – Cross-document [new]: Item 1A risk paragraph ↔ earnings call

passage, same firm, same period,

TF-IDF matched

Hard negative: call passage from a different sector

This forces the encoder to learn that risk language is consistent across document registers such as formal SEC disclosure versus verbal CFO Q&A. HTML (Yang, Ng, Smyth & Dong, 2020, WWW 2020) and VoTAGE (Sawhney et al., EMNLP 2020) establish empirically that calls and filings carry complementary but overlapping risk information. The extension of Chiu et al.'s (EMNLP 2025) dual-view framework to a cross-document four-view architecture is, to my knowledge, not present in the existing literature, which positions it as the novel methodological contribution of the project.

Item 7A: Conditionally Recommended for DAPT Corpus Expansion

Item 7A ("Quantitative and Qualitative Disclosures About Market Risk") discusses interest rate exposure, FX risk, and credit risk in explicit financial terms which are denser in quantitative risk vocabulary than Item 1A. Jehnen et al. (2026, FintextSim) explicitly train on Item 7 + Item 7A alongside Item 1A and achieve their best domain-adaptation results which is a direct published precedent for this exact corpus expansion. Extracting Item 7A from the collected files would require approximately one day of work (regex-based section extractor). The marginal gain over Item 1A alone is uncertain, but the precedent is clear indication to incorporate Item 7A.

I would welcome the opportunity to discuss these decisions in a brief meeting if you are available, particularly the 4th contrastive view and the scope of earnings call integration, as both have a direct bearing on the Phase 2 implementation plan. I am also happy to address any questions you may have at your convenience.

Thank you for your continued guidance.

Best regards,

Sarthak Govind Vishnu

MSc Artificial Intelligence

From: Sarthak Vishnu <S.G.Vishnu@sms.ed.ac.uk>

Sent: 10 June 2026 11:19

To: Tiejun Ma <tiejun.ma@ed.ac.uk>; Sean Choi <schoi3@ed.ac.uk>

Cc: Sunnie Li <sunnie.y.li@ed.ac.uk>

Subject: Re: Data Coverage on Finstream Server for 10-K/10-Q and Earnings Calls

Dear Prof Ma,

Thank you for the guidance.

To confirm, I have already collected Form 4 insider trading data (2006– 2026 Q1) and am currently working on GDELT - Events DB.

For a quick status update on all datasets discussed, the Form 4 (2006-2026 Q1), 10K & 10Q (2025-2026), and Earning Call Transcripts (2025-2026) are collected. Next, I will prioritise completing the GDELT collection and in parallel review relevant literature to identify where and how each of these datasets can best be incorporated into my pipeline.

Best regards,
Sarthak

From: Tiejun Ma <tiejun.ma@ed.ac.uk>
Sent: 10 June 2026 10:57
To: Sarthak Vishnu <S.G.Vishnu@sms.ed.ac.uk>; Sean Choi <schoi3@ed.ac.uk>
Cc: Sunnie Li <sunnie.y.li@ed.ac.uk>
Subject: Re: Data Coverage on Finstream Server for 10-K/10-Q and Earnings Calls

Hi Sarthak,

I suggest that you start with the smaller datasets and process your development work first. Have you got the GDELT, Form 4 etc datasets? Please priority this part of the data collection first.

We may discuss about long history data later once you have shown progress. It will also reduce the workload for the server as well

Best wishes
Tiejun

From: Sarthak Vishnu <S.G.Vishnu@sms.ed.ac.uk>
Sent: Wednesday, June 10, 2026 10:50 am
To: Sean Choi <schoi3@ed.ac.uk>
Cc: Tiejun Ma <tiejun.ma@ed.ac.uk>; Sunnie Li <sunnie.y.li@ed.ac.uk>
Subject: Data Coverage on Finstream Server for 10-K/10-Q and Earnings Calls

Hi Sean,

I have downloaded the **10-K/10-Q filings** from `/data/finstream/airflow/data/SP500/sec` and **earnings call transcripts** from `/data/finstream/airflow/data/SP500/calls` on the Finstream server. However, I noticed that both datasets only cover 2025–2026, while the Data Document states the earnings calls should span 2008–2024 and the SEC filings 2006–2024.

For my dissertation, I need data from 2006 to 2026. Could you clarify:

1. Is the historical 10-K/10-Q data (2006–2024) stored elsewhere on the server, or at a different path I may have missed?
2. Similarly, the earnings calls downloaded from that path only contain 2025–2026 data (172 tickers). Is the full 2008–2024 archive (37,962 transcripts) stored in a different location, perhaps in MongoDB or a separate storage volume?
3. If the historical data is not accessible as flat files, should I collect the 10-K/10-Q filings from SEC EDGAR myself, and the earnings calls via API Ninjas directly?

Any guidance before I start a potentially large collection run would be much appreciated.

Best regards,
SarthaK